P9 — Error log of PROJECT spacetimes

Provenance: every error made during the work, how it was found, and how it was corrected

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Purpose

Hard Rule 7 of the task specification forbids shortcuts and demands that problems be explained. This file therefore records every error that was made while the project was carried out, including errors that were corrected before anything was published. For each error it states what was wrong, how it was detected, what was done, and where the corrected statement can be checked. An error that is not listed here and is found later is to be added, not silently repaired.

Errors in the construction and verification of the Dirac matrices

No.	What was wrong	How it was found	Correction	Where to check
E1	The first version of check `71-P4` stated the block form of the complex projectors of 7+1 with the wrong sign of the off-diagonal blocks.	The program compares the statement with the computed matrices; the check failed.	The statement was corrected: χ_{1671} has the blocks $(0, -i;\ i, 0)$, and $2P^{L}_{1671}$ has the blocks $(1, i;\ -i, 1)$.	`data/matrix_checks.txt`, line `71-P4`
E2	The 7+1 matrices of the project were first defined as the covariant matrices `Γ17[swap04(A)]` of `7+1.nb`. The Dirac equation of the notebook, however, is written with the contravariant matrices `Γ17up`.	Re-reading sections 5.3 and 7.1 of `7+1.nb` while writing the Dirac equation of dirac1671.	The project's matrices are $\gamma^{A}_{1671} =$ `Γ17up[swap04(A)]`; then ω_{1671} equals `Γ17[8]` exactly and χ_{1671} equals `χ17`.	`data/matrix_checks.txt`, lines `71-N2`, `71-V1`, `71-P1`
E3	Check `71-Q9` first claimed that the two-particle content of $\psi^{T} C^{-}_{1671} \gamma^{j}_{1671} \psi$ is $+\tau^{j}_{1671}$.	The check failed.	The content is $-\tau^{j}_{1671}$.	line `71-Q9`
E4	Check `71-P5` first claimed that the unitary matrix U^{c}_{1671} maps P^{L}_{1671} to P^{L}_{1644}.	The check failed.	It maps P^{L}_{1671} to P^{R}_{1644} and P^{R}_{1671} to P^{L}_{1644}; the reason (two different sign conventions for the chirality matrix in `7+1.nb`) is stated in the check and in Sec. 2.6 of "dirac matrices".	line `71-P5`

No.	What was wrong	How it was found	Correction	Where to check
E5	A commit was created although one check (E4) still failed, because the commands were chained in a way that did not stop on failure; its message claimed that all checks pass.	Reading the output after the commit.	The check was corrected, the program was run again (all checks passed), and the commit was amended before it left the workspace; it was never pushed in the faulty state. Since then a commit is made only if the check program returns success.	`git log`: commit "Dirac equations (dirac1644, dirac1671) and Pauli classification of fermion pairs"
E6	The first implementation of the exact rank (fraction-free elimination with 128-bit integers) overflowed for 256×256 matrices.	The Rust program is compiled with overflow checks and stopped with an error instead of returning a wrong number.	Ranks are computed modulo the prime $2^{61}-1$, with the argument (in the documentation of `rank`) that a rank found modulo a prime is a lower bound of the rank over the rational numbers, so that a full rank is proved; statements about a null space are in addition backed by an explicit null vector. The Python script confirms the same statements by other methods (Gram matrices, enumeration, exact rational elimination).	`code/spacetimes-rs/src/imat.rs`; `data/matrix_checks_independent.txt`
E7	**Symbol collision introduced by this project.** Until November 12, 2026 the program and the drafts used the names `g44`, `g71`, `eta44`, `eta71`, `gamma44`, `gamma71`, `chi71` and the like for the project's matrices. The author's notebooks use these very names for other objects: `g44` and `g71` are metric tensors of the gravity notebooks, `η44` is a 4×4 metric in `Dirac_Nash_Identity.nb`, and `η71`, `Γ71`, `χ71`, `PL71`, `PR71` of `7+1.nb` differ from the project's 7+1 metric, matrices, chirality matrix and projectors. This is exactly the kind of ambiguity that Hard Rule 5 forbids.	A systematic search of all notebooks for names carrying the tags 44, 71 and 17, made while the symbol table of "dirac matrices" was written.	All project symbols were renamed with the tags 1644 and 1671, which occur in no original file (source code, check statements, data files, documents). Other letters with two meanings were separated as well: the 8×8 matrix σ became σ_{88}, the permutation "π" became swap_{04}, the 8×8 blocks "ρ" of 7+1 became τ^{A}_{1671}, the multi-index "K" was replaced by explicit index sets, and rotation parameters are written θ_{AB} instead of ω_{AB}.	MASTER provenance P1; Sec. 0.2 of "dirac matrices"
E8	In the first draft of "dirac matrices" the proof of Lorentz covariance used the transformation $\psi'(x) = S^{-1}\psi(\Lambda x)$ and contracted the indices of Λ incorrectly.	Re-deriving every formula of the draft before the document was assembled.	The correct statement is $\psi'(x) = S\,\psi(\Lambda x)$ with $S\gamma^{A}S^{-1} = \Lambda^{A}{}_{B}\gamma^{B}$; the derivation is given in Sec. 1.8.2. The faulty draft was never committed.	"dirac matrices", Sec. 1.8.2
E9	The first version of the routine of the Python script that counts invariant pair matrices mixed the linear equations of different generators.	Review of the code before its first complete run.	Each generator contributes its own 64 equations for the 28 unknowns. The corrected routine reproduces the numbers of the Rust program (0 and 10 for 7+1; 6, 10, 1 and 0 for 4+4). The faulty version was never committed.	`code/scripts/verify_matrices_independent.py`, function `invariant_dimension`

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E10	The draft of Sec. 1.6 said that cell 166 of `4+4.nb` verifies all five projector relations; the cell verifies three of them.	Comparing the sentence with the cell.	The text now says which three relations the notebook verifies; all five are verified by checks `44-P1`, `44-P2`. Never committed in the faulty form.	“dirac matrices”, Sec. 1.6

Errors in the pairing calculations

No.	What was wrong	How it was found	Correction	Where to check
E11	A formula derived by hand for the exchange amplitude of the radial mode in the two-form pair channels, proportional to $4m^2 - q^2$, was wrong.	The program `spacetimes channels` computes every amplitude numerically from the Dirac matrices; the hand formula disagreed with it.	The amplitude of the radial mode is the same in the vector and in the two-form channels, $(4m^2 + q^2)/4E^2$ on the mass shell; only the phase mode distinguishes the channels. The hand formula was never written into a document.	`data/pair_channel_kernel.csv`; unit test `kernel::tests::hidden_axis_closed_forms`
E12	In the unit test of the angular average of the pairing kernel (`gap.rs`), the reference value was computed with a 64-point Gauss rule, which is inaccurate for the nearly singular integrand; the test failed although the analytic formula was right.	The test failed; refining the reference quadrature made it converge to the analytic value.	The reference is now a composite quadrature graded towards the singularity.	unit test `gap::tests::angular_average_matches_quadrature`
E13	Documentation comments of `gap.rs` and `kernel.rs` contain indented formulas, which the Rust test tool treats as code examples and tries to compile. Commit `97df13f` was made after `cargo test` reported success for the unit tests; the documentation-test stage, which runs afterwards, failed and was overlooked.	Running the complete test command again before the next commit.	The crate declares `doctest = false` (the formulas are not code). All unit tests passed in `97df13f`; no result is affected.	`code/spacetimes-rs/Cargo.toml`
E14	`dimer.rs`: the number of bound states was taken to be the number of nodes of the zero-energy solution inside the integration range; the node that lies beyond the range just above threshold was not counted, which gave the critical coupling $D_c = 1.718$ instead of 1.6798.	The unit test compares D_c with the literature value and failed.	The node of the asymptotic straight line is counted ($u\,u' < 0$ at the end of the range). Never committed in the faulty form.	unit test `dimer::tests::critical_coupling`; `data/dimer.csv`

Errors in the cosmological programs

All errors in this section were found by comparing the Rust solver with the code CLASS (spectra of the author’s NCSS project) *before* any cosmological result was committed; none of them reached a document. They are recorded because the comparison is what makes the final agreement (Supplement S3, Sec. 2) meaningful.

No.	What was wrong	How it was found	Correction	Where to check
E15	`spectra.rs`: wavenumbers above $\max(2.45\,\ell/\tau_0,\ 0.012\ \mathrm{Mpc}^{-1})$ were not allowed to contribute to the multipole ℓ. The projection of a wave of wavenumber k contributes to all $\ell < k\tau_0$ with a slowly decaying weight, so power was lost: the temperature spectrum was too low by up to 14 per cent at $\ell = 100$.	Comparison with CLASS.	The cut was removed; only the cut $\ell > 1.02\,k\tau_0 + 10$ (where the spherical Bessel function is negligible) remains.	`spectra.rs`, comment in `compute`
E16	`bessel.rs`: a unit test demanded an interpolation error of the Bessel table below $3 \times 10^{-7}/x$, which is stricter than the error bound $(h^4/384)\max\lvert f''''\rvert$ of cubic Hermite interpolation with the table's step $h = 0.2$. The code was right and the test was wrong.	The test failed; the measured error was compared with the theoretical bound.	The test now checks the theoretical bounds of value and derivative, and a second test checks the differential equation.	unit tests `bessel::tests::table_interpolation_error_bounds`, `differential_equation`
E17	`boltzmann.rs`: for the matter power spectrum at large wavenumbers the perturbations of the radiation were simply dropped at late times. In the synchronous gauge this is inconsistent: the frame moves with the dark matter, so the radiation has a bulk velocity $-k^2\alpha$ relative to it, and its momentum density is not small compared with that of the baryons. The matter power spectrum was 11 per cent too high for $k > 0.33\ \mathrm{Mpc}^{-1}$.	Comparison of $P(k)$ with CLASS.	Replaced by the radiation-streaming approximation, which keeps these terms. The fast and the full integration now agree to 5×10^{-5} ($k \le 0.42$) and 3×10^{-4} ($k = 2\ \mathrm{Mpc}^{-1}$).	unit test `boltzmann::pk_tests::matter_only_mode_agrees_with_full_integration`
E18	`cosmo.rs`: recombination was computed with the equations of RECFAST 1.0 (factor 1.14 for hydrogen, singlet channel only for helium), which lack the corrections that reproduce the modern multi-level calculations. Effect: 100 θ_s off by 3.4×10^{-4} (relative), spectra off by up to 1.5 per cent at $\ell = 2000$.	Comparison with CLASS (which uses HyRec): a smooth, growing difference in TT and EE after lensing had been accounted for.	The equations of RECFAST 1.5.2 are used: factor 1.125, the two Gaussian corrections of the Lyman-alpha escape factor, and for helium the modified escape probability with hydrogen continuum opacity and the triplet channel. 100 θ_s and r_{drag} now agree with CLASS to 1.3×10^{-5} and 3×10^{-5}.	`cosmo.rs`, `Thermo::new`; unit test `cosmo::tests::background_and_recombination_of_the_fiducial_model`
E19	`cosmo.rs`: while the gas is tightly coupled to the radiation, the temperature equation is extremely stiff. The Runge-Kutta stages advanced the temperature with this stiff derivative; at $z = 2812$, in the first step of the helium rate equation, a stage produced a negative temperature and a NaN, and the program then set the fraction of singly ionized helium to zero. Helium "recombined" within one step instead of gradually between $z = 3000$ and $z = 1700$.	After E18 a difference of 1 per cent at $\ell = 2000$ remained; switching the helium corrections on and off changed nothing at all, which exposed the fault; a printout of $x_e(z)$ confirmed it.	Every stage uses the quasi-static solution of the temperature equation while the coupling is tight (as RECFAST does); the program stops if an ionization fraction is not finite.	unit test `cosmo::tests::ionization_history` (fails for the faulty version)

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E20	`validate_cmb_against_class.py`: the Rust matter power spectrum was interpolated linearly in the logarithms onto the wavenumbers of the reference; near the maximum of $P(k)$, where the Rust grid is coarse, this produced an apparent difference of 1.3 per cent.	The difference was confined to the region where the grid spacing changes.	Cubic-spline interpolation; the difference is 1×10^{-4} there.	`data/pk_validation.csv`
E21	The fiducial amplitude was $A_s = e^{3.0445} \times 10^{-10}$, while the reference spectra of the NCSS project were computed with $A_s = 2.1 \times 10^{-9}$ ($\ln(10^{10} A_s) = 3.04452$): a difference of 2×10^{-5}.	Reading the header of the reference file.	The fiducial model uses $A_s = 2.1 \times 10^{-9}$.	`cosmo.rs`, `Params::fiducial`
E22	A first run of the mock-data fits (`spacetimes forecast`) was started before the validation against CLASS was complete, with the faults E18 and E19 present.	—	The run was stopped and its output discarded; all results in the repository come from the corrected program.	commit `045e161` and later

Errors in drafts of the documents

No.	What was wrong	How it was found	Correction	Where to check
E23	The draft of Supplement S1 stated a closed formula, $F_P = [q^2 - 2(\vec{q} \cdot \hat{n})^2]/4E^2$, for the phase-mode amplitude of a vector pair polarized along an axis $\hat{n}$ of ordinary space, and concluded that its angular average is one third of that of an extra-axis pair.	Before the text was accepted, the formula was compared with `data/pair_channel_kernel.csv`; it is wrong for pairs polarized in ordinary space (their two-particle content depends on the direction of the momentum).	The formula and the conclusion were removed; the text states only what the data show. Never committed in the faulty form.	S1, Sec. 4.2 (iii)
E24	The draft of S1 conjectured that the quartic Ginzburg–Landau invariant is $2(\sum \lvert\Delta^a\rvert^2)^2 - \lvert\sum_a (\Delta^a)^2\rvert^2$ in both spacetimes.	The machine check failed for 4+4 (deviation 4).	The correct invariant contains the metric of the spacetime on the four extra axes, $\lvert\eta_{ab}\Delta^a\Delta^b\rvert^2$; in 4+4 the components along the time-like axes enter with the opposite sign. Never committed in the faulty form.	S1, Eq. (6.2); `data/gl_quartic_check.csv`
E25	The draft of S1 gave the Ginzburg–Landau coefficients a, b and the stiffness K with factors of 2 that did not match the stated normalization of the energy.	Checking that the uniform solution reproduces the textbook result $\lvert\Delta\rvert^2 = \frac{8\pi^2}{7\zeta(3)} T_c^2 (1 - T/T_c)$.	Coefficients corrected: $a = \frac{1}{2} N(0) \ln(T/T_c)$, $b = 7\zeta(3) N(0)/16\pi^2 T_c^2$, $K = 7\zeta(3) N(0) v_F^2/96\pi^2 T_c^2$. Never committed in the faulty form.	S1, Secs. 6.2 and 7

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E26	The draft of S1 (Sec. 6.3) stated that in 4+4 the degeneracy between the hidden-axis singlet and the triplet of pairs along the additional times is *accidental*, "no symmetry enforces it", and that terms of higher order would lift it; the draft of S2 concluded from this that the 4+4 condensate is a singlet under every internal symmetry.	While the order-parameter space was prepared for the checks with the effective action of Cornwall, Jackiw and Tomboulis (S4), the set of all matrices that commute with the dynamics of the zero-mode sector was computed (`symmetry.rs`). It has dimension 8 in both signatures: in 4+4 the matrices $\gamma^0\gamma^t$ are Hermitian and generate, with the rotations of the additional times, a hidden $SO(4)$ that rotates the singlet into the triplet.	The degeneracy is exact for fields independent of the extra coordinates and is lifted only by a dependence on $x0$. S1 (new Sec. 2.1, Secs. 6.3 and 9) and S2 (Sec. 4.2 and the verdict) were rewritten. Never committed in the faulty form.	`data/reduced_symmetry.csv`; unit test `symmetry::tests::zero_mode_symmetry_is_u2_times_u2_in_both_signatures`
E27	`cjt_normal.rs`, finite temperature: the thermal part of the fermion loop was kept and its vacuum part dropped. Each of the two contains a term $(g\phi)^4 \ln(g\phi)$ that is not analytic at $\phi = 0$; they cancel only in the sum. With the thermal part alone the potential had a spurious minimum at small ϕ, and the expectation value appeared to fall to zero in a long smooth tail.	The tail did not look like any known behaviour of the transition; the small-ϕ expansion of the thermal integral showed the logarithm.	The renormalized vacuum loop (the function `v_sea`, fixed by the two conditions that the symmetry allows) is included. The closed estimates of the transition temperature acquire the term $\nu g^2 m^2/8\pi^2$. Never committed in the faulty form.	`cjt_normal.rs`, documentation of `Thermal`; `data/cjt_thermal_profile.csv`
E28	The draft of S2 (Sec. 3) estimated the coupling needed for a pair condensate in the vacuum from the gap equation of S1 with particles only, and found that it would require $\ln(2\Lambda/m) \approx 1 + 16\pi^2/g^2$. Without a Fermi surface the negative-energy states contribute as much as the positive-energy ones, and the momentum dependence of the self-energy is essential; the estimate was not valid.	The vacuum problem was set up properly as the stationarity condition of the CJT action (ladder equation, `cjt_vacuum.rs`).	The critical coupling is $\alpha_c \to \pi/2$ for a large cutoff and between 2.2 and 4.2 for cutoffs of 100 to 10 fermion masses; S2 quotes these values. Never committed in the faulty form.	`data/cjt_vacuum.csv`; unit test `cjt_vacuum::tests::scale_invariant_ladder_equation`
E29	In the first drawing of the figure of the matter power spectrum of superfluid dark matter the legend read "$m = 05$ keV" and "$c_{s0} = 3 \times 10^{13} c$": the plotting library removes points and minus signs from column names, and the labels had been rebuilt from the names.	Looking at the figure.	The labels are taken from the numerical columns of `data/sfdm_summary.csv`. Never committed in the faulty form.	`make_figures.py`, `fig_sfdm`
E30	The first complete run of `spacetimes forecast` was made with a program built before the last change of the initial conditions of the dark fluid (velocity perturbation for an equation of state different from 1/3). The fits of the stepped fluids from that run differ from those of the committed program by up to 4×10^{-4} in χ^2 and 7×10^{-4} km s^{-1} Mpc^{-1} in H_0.	A second run, made to add a seventh family, was compared with the first.	All data in the repository are from the second run, made with the committed program.	`data/dr_profiles.csv`

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E31	The first comparison of the Ginzburg–Landau coefficients extracted from the CJT potential with the formulas of S1 used amplitudes up to $0.65\,T_c$ and found the quartic coefficient off by up to 15 per cent away from T_c. Two effects were mixed: terms beyond the sixth order at those amplitudes, and a genuine factor $1 + 4(\eta(T) - 1)$ of the potential along a ray of fixed shape, which equals 1 only at T_c.	The deviation was proportional to $\ln(T/T_c)$ with a coefficient equal to four times the coupling.	Smaller amplitudes; the coefficients are compared at T_c, where they agree with S1 within 1 per cent; the factor is derived in S4.	`data/cjt_gl.csv`, `data/cjt_gl_summary.csv`
E32	`cjt_normal.rs`, finite temperature: the four thermal integrals were taken from a table by *linear* interpolation. A piecewise linear function has a discontinuous derivative at every node of the table, and the Hartree equations use derivatives of the potential with respect to the expectation value (one of them by finite differences over a step much smaller than the spacing of the table). The position of the minimum therefore jumped from node to node as the temperature changed. The melting temperatures were affected in the third digit; the mass of the phase mode, a small difference, was visibly noisy for the weakest coupling ($g_{VS} = 0.1$: scatter of 30 per cent). The faulty table `data/cjt_thermal_profile.csv` is in commit `489d7f6`; no document quoted it.	Looking at the figure of the phase-mode mass against temperature: one curve was jagged.	Cubic interpolation through the four nearest points of the table (relative error below 10^{-8}, against 10^{-6} before, and a continuous derivative to the accuracy needed). The curves are smooth; $T_{1/2}/v_{VS}$ of the weakest coupling changed from 0.679 to 0.677.	`cjt_normal.rs`, `thermal_fast`; unit test `cjt_normal::tests::thermal_table_interpolation`; `figures/fig_cjt_thermal.png`
E33	The draft of S4 contained a table of the internal symmetry whose entries were copied from the data file with vertical bars (for norms) and carets in them. A vertical bar separates the cells of a Markdown table, so several rows were split into the wrong columns in the PDF, although the build reported no error.	Reading the PDF after the build.	The entries are typeset by `make_tables_cjt.py` (`_symmetry_item`, `_symmetry_value`), which stops with an error if a vertical bar would reach a table. Never committed in the faulty form.	S4, Sec. 2.4
E34	The draft of S4 (Sec. 1.7) said that the truncations of Γ_2 "with an odd number of terms" have no solution of the gap equation at strong coupling. It is the truncations at three and at five loops (two and four terms), whose last term in the self-energy is negative. The draft of Sec. 6 also stated that a discontinuous thermal transition "is expected on physical grounds" for a large Yukawa coupling; the calculation does not support a statement about the order of the transition either way.	Comparing each sentence of the draft with the tables it refers to.	Both sentences corrected. Never committed in the faulty form.	S4, Secs. 1.7 and 6

No.	What was wrong	How it was found	Correction	Where to check
E35	In the output of the program (`data/matrix_checks.*`, checks `44-Q1` and `44-Q2`; `data/pair_classification.*`) the two $SU(2)$ factors of the rotation group of the four space coordinates $x0, \ldots, x3$ of the 4+4 spacetime were called A and B — the same letters that the 7+1 lines, and later Supplements S1, S2 and S4, use for the two factors of the internal symmetry of the fields that do not depend on the extra coordinates. In 4+4 these are two different groups (the first contains the ordinary rotations, the second does not act on ordinary space). No statement was wrong, but the same symbol stood for two objects, which Hard Rule 5 forbids. In commits up to `b6c6736`.	Found while the manuscripts were written, when both groups had to appear in one section.	The space-rotation factors of 4+4 are now called X and Y everywhere (check texts, the table `pair_classification.csv` has separate columns `dim_X`, `dim_Y`); "dirac matrices", Sec. 3.2 and Appendix A, says so.	`pairs.rs`; P1 (MASTER provenance), list of the project's own symbols; the two data files of the pair classification